

Report on Enhancing Mixed Traffic Flow with Platoon Control and Lane Management for Connected and Autonomous Vehicles

Abstract—
Index Terms—

I. INTRODUCTION

The rapid advancement of Connected and Autonomous Vehicles (CAVs) is reshaping the landscape of modern transportation. As these intelligent systems integrate into existing traffic environments dominated by Human-Driven Vehicles (HDVs), optimizing their coexistence becomes a critical challenge. Despite the potential of CAVs to enhance road safety, traffic efficiency, and fuel consumption, their effectiveness in mixed traffic scenarios remains uncertain. Issues such as non-uniform driving behaviors, unpredictable human responses, and varying levels of connectivity introduce complexities that demand strategic control solutions.

One of the key challenges in mixed traffic is ensuring smooth traffic flow while mitigating congestion and instability. Traditional traffic control models often fail to accommodate the dynamic interactions between CAVs and HDVs, leading to inefficiencies, particularly at merging zones and high-density road segments. Moreover, without an intelligent coordination mechanism, CAVs may not fully leverage their communication capabilities, limiting their potential to optimize mobility.

To address these challenges, this study proposes a hybrid control framework that integrates platoon-based coordination with dedicated lane management to enhance traffic efficiency in mixed environments. This approach enables CAVs to dynamically form and maintain platoons while adapting their behavior based on surrounding HDVs and road conditions. By implementing cooperative adaptive cruise control (CACC) strategies, the system aims to reduce disruptions, improve lane discipline, and ensure smoother vehicle transitions at critical points like highway on-ramps.

The methodology of this study is structured around microscopic traffic simulation using SUMO, providing a detailed evaluation of the hybrid framework's impact under varying levels of CAV market penetration. Key performance indicators such as speed, traffic density, and congestion levels are analyzed to assess the benefits of intelligent platoon formation and lane coordination. The results offer insights into how emerging CAV technologies can be strategically

deployed to enhance road network performance, reduce bottlenecks, and facilitate seamless integration of autonomous systems into real-world traffic.

This paper is organized as follows: Section II outlines the methodology detailing the simulation setup, control models, and evaluation criteria. Section III presents the results and discussion, highlighting the performance improvements achieved through the hybrid control strategy. Finally, Section IV concludes with key findings, policy implications, and potential future research directions.

II. METHODOLOGY

A. Overview

This study uses microscopic traffic simulation in SUMO to analyze the effect of Connected and Autonomous Vehicles (CAVs) and Human-Driven Vehicles (HDVs) on mixed traffic flow. A hybrid control framework was proposed that combines platoon-based vehicle coordination with dedicated lane management. The methodology is divided into simulation setup, model descriptions, detailed explanation of the platoon control mechanism, and parameter utilization.

B. Simulation Setup

The simulation was carried out using a three-lane highway network with an on-ramp merging section, modeled over a total length of 3.25 km. The acceleration lane for merging traffic is 250 meters long. The network structure was defined in the SUMO network file with edges and lanes having speeds of 30.56 m/s (approximately 110 km/h) on the main road and 22.22 m/s (80 km/h) on the ramp. Each lane has a width of 3.5 meters. Traffic flows originate from two main routes: Route 1 (edges 1, 3, 4) for the mainline traffic and Route 2 (edges 2, 3, 4) for ramp vehicles.

Two vehicle types were defined: Human-Driven Vehicles (HDVs) and Connected and Autonomous Vehicles (CAVs). HDVs have a length of 5 meters, acceleration of 1.5 m/s², deceleration of 3.5 m/s², emergency deceleration of 3.5 m/s², minimum gap of 3.5 meters, and a driver reaction time (τ) of 2.0 seconds. They are visually represented in the simulation with a green color (RGB 127,255,0). CAVs are defined with improved acceleration of 2.6 m/s², deceleration of 4.5 m/s², emergency deceleration of 4.5 m/s², minimum gap of 2.0

meters, and a shorter tau of 1.1 seconds. CAVs are colored red (RGB 255,0,0) for clear differentiation.

The simulation time was set to 3600 seconds with a warm-up phase of 300 seconds and a simulation step interval of 0.1 seconds. Vehicle generation was controlled through predefined flows with a total of 4,320 HDVs and 1,080 CAVs per hour on the main route (Route 1) and 960 HDVs with 240 CAVs per hour on the ramp route (Route 2). All vehicles departed with the maximum allowed speed and random lane assignment to reflect realistic conditions. The “lcKeepRight” parameter was disabled to allow vehicles to dynamically change lanes based on control algorithms and traffic conditions.

Traffic detectors were placed along the stretch to monitor parameters such as vehicle speed, flow, and density at multiple checkpoints. The Market Penetration Rate (MPR) of CAVs was varied from 0% to 100% in increments of 20% to evaluate the impact of connected vehicle integration. The network also incorporated junctions with zipper merging logic, ensuring a realistic simulation of merging behavior at the on-ramp.

Overall, this setup provides a controlled yet dynamic environment for testing hybrid control strategies and observing their effectiveness in mitigating congestion, especially at critical merging zones beyond the 2000-meter mark.

C. Intelligent Driver Model (IDM) for HDVs

The IDM governs HDV behavior using the following formula:

$$a = a_m \left[1 - \left(\frac{v}{v_d} \right)^4 - \left(\frac{s^*(v, \Delta v)}{s} \right)^2 \right] \quad (1)$$

where:

- a_m = Maximum desired acceleration (1.4 m/s²)
- v = Current speed
- v_d = Desired speed (110 km/h or 30.56 m/s)
- s = Actual gap
- $s^*(v, \Delta v) = s_0 + \max(0, vT + \frac{v\Delta v}{2\sqrt{a_m b_m}})$
- s_0 = Minimum gap (2 m), T = safe headway (1.5 s), b_m = comfortable deceleration (2 m/s²)

D. Cooperative Adaptive Cruise Control (CACC) for CAVs

CAVs operate based on catch-up and following modes. Acceleration is controlled using:

$$a_{i,t} = k_p e_{i,t-1} + k_d \dot{e}_{i,t-1} \quad (2)$$

where:

- $e_{i,t} = x_{i-1,t} - x_{i,t} - d - Tv_{i,t}$ (desired gap error)
- Gains: $k_p = 0.5$, $k_d = 0.3$

When no leader vehicle is within range, speed control mode is applied using:

$$a_{i,t} = k(v_d - v_{i,t-1}) \quad (3)$$

with $k = 0.4$.

E. Detailed Explanation of the Platoon Control Strategy

The platoon control strategy begins with each CAV assessing the presence and position of nearby vehicles. If a CAV identifies a leader vehicle and the distance between them is less than 20 meters, it immediately enters the following mode. In this mode, the vehicle adopts a reduced time headway of 0.6 seconds and a speed factor of 1.1, enabling it to maintain close proximity to the leader and respond quickly to any changes.

If the relative distance between the following CAV and the leader ranges between 20 meters and 120 meters, the vehicle transitions into the catch-up mode. Here, the system increases the desired speed using a speed factor of 1.2 and a time headway of 0.7 seconds. This adjustment allows the vehicle to safely and rapidly close the gap without causing instability.

In cases where the following CAV is approaching a platoon but not yet within the immediate following threshold, it enters the lead-catching mode. This mode also uses a speed factor of 1.2 and a time headway of 0.7 seconds, allowing the vehicle to adjust its velocity to synchronize with the platoon leader ahead.

If a vehicle is within the catch-up range but the front platoon is nearing its maximum capacity (nine vehicles), the vehicle will adopt the catch-up following mode. In this state, the vehicle uses an even higher speed factor of 1.3 and a headway of 0.6 seconds, helping it position itself optimally behind the leader while maintaining platoon stability.

The leader vehicle, which is the frontmost in the platoon, operates in the leading mode. This mode uses a larger headway of 1.1 seconds and a speed factor of 1.0 to ensure smooth movement and safety for all following vehicles. These dynamic adjustments in time headway and speed factor ensure that the platoon remains compact and stable while allowing smooth formation and disbanding in mixed traffic conditions.

III. RESULT AND ANALYSIS

The spatio-temporal heatmaps presented show traffic conditions in two distinct scenarios: the baseline scenario without platoon or lane control, and the hybrid control scenario where both platoon management and lane control are applied. The road setup includes a three-lane highway with a merging area after 2000 meters.

In the baseline scenario, significant congestion is observed at the merging area after the 2000-meter mark. This is indicated by large red patches on the heatmaps, signifying lower speeds and traffic accumulation. With an increase in Market Penetration Rate (MPR), congestion intensity slightly decreases, but bottlenecks remain prominent, particularly in

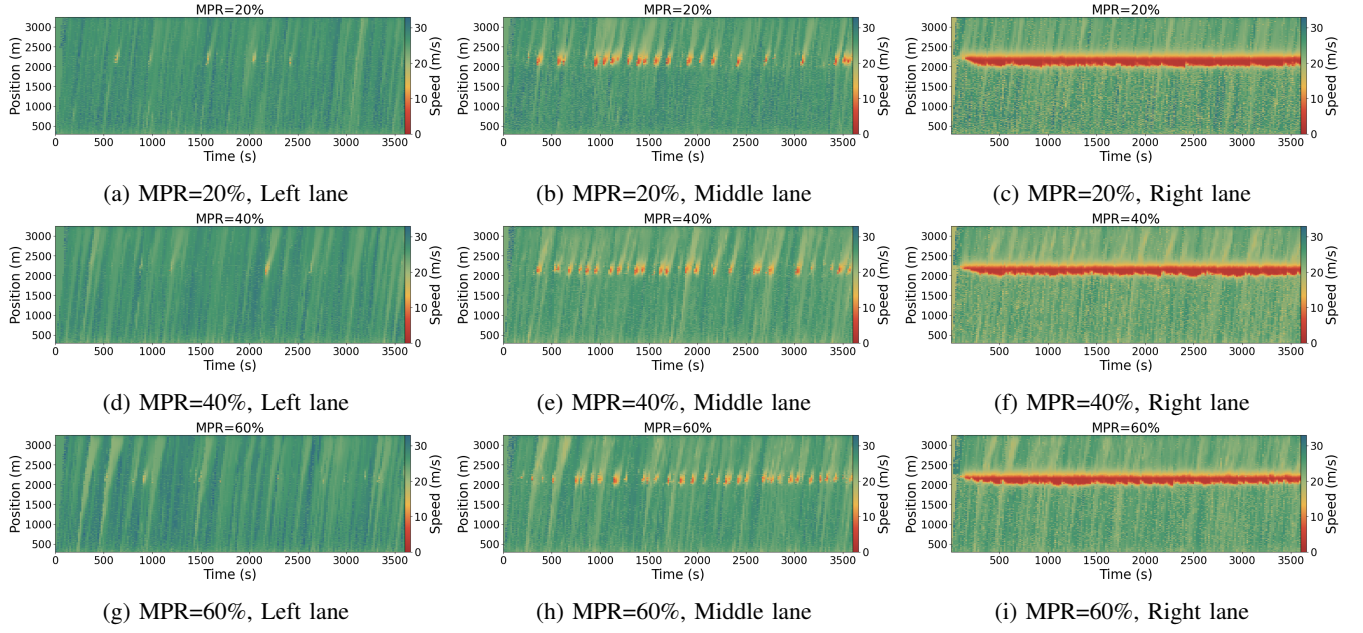


Fig. 1: Spatio-temporal heatmaps for different Market Penetration Rates (MPRs) in baseline scenario for three lanes: (a) Left lane, (b) Middle lane, and (c) Right lane.

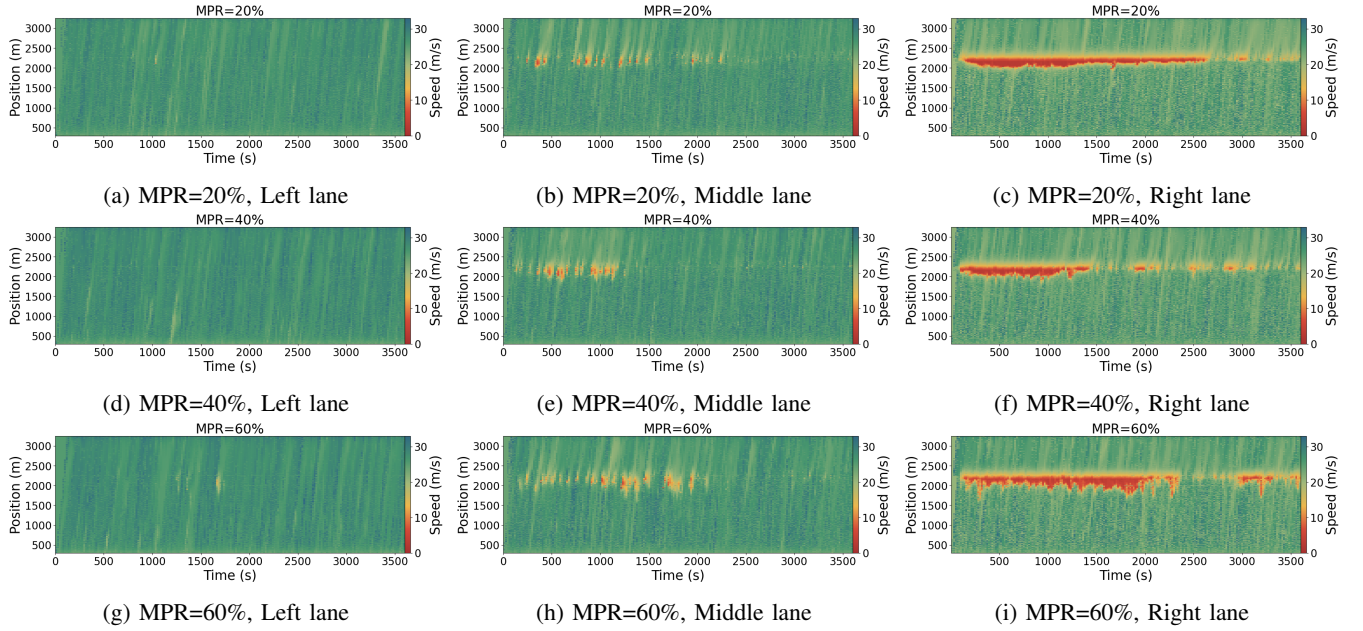


Fig. 2: Spatio-temporal heatmaps for different Market Penetration Rates (MPRs) in hybrid control scenario for three lanes: (a) Left lane, (b) Middle lane, and (c) Right lane.

the right lane where merging occurs.

On the other hand, the hybrid control scenario demonstrates considerable improvement as the MPR increases. The application of platoon-based control and dedicated lane management allows smoother merging and better lane utilization. At lower MPRs (20%), minor congestion still occurs; however, as the MPR reaches 40% and beyond, the

heatmaps show significantly fewer red zones. This indicates that vehicles are able to maintain higher speeds and smoother flow, even in the merging sections.

The left and middle lanes in the hybrid scenario remain relatively free-flowing with increased MPR, while the right lane—previously a congestion hotspot—also experiences substantial improvement. This confirms that the hybrid control

strategy, with proper platoon formation and lane management, helps alleviate congestion issues that typically arise from merging maneuvers.

IV. CONCLUSION

This study has explored the effectiveness of a hybrid traffic control framework that integrates platoon coordination and dedicated lane management for mixed traffic environments with both Connected and Autonomous Vehicles (CAVs) and Human-Driven Vehicles (HDVs). Through detailed simulation analysis using SUMO, it has been demonstrated that the proposed strategy significantly reduces congestion, particularly in areas prone to traffic buildup, such as merging zones.

The findings reveal that, under baseline conditions without platoon or lane control, severe bottlenecks occur, especially at the on-ramp merging sections. However, with the application of hybrid control and increasing Market Penetration Rates (MPR) of CAVs, traffic flow improves noticeably. Higher MPRs enable more stable platoon formations and smoother merging behaviors, resulting in reduced stop-and-go waves and higher average speeds.

The spatio-temporal analysis clearly indicates that the hybrid control strategy not only minimizes congestion in the right lanes but also ensures uniform traffic distribution across all lanes. As CAV adoption continues to rise, such control approaches can play a pivotal role in transforming current traffic systems into safer and more efficient networks.

In conclusion, this research confirms that implementing platoon control and lane management strategies for CAVs in mixed traffic scenarios can greatly enhance roadway performance. Future work may focus on real-world validation, adaptive algorithms that respond to fluctuating traffic demands, and integration with broader smart city traffic systems.